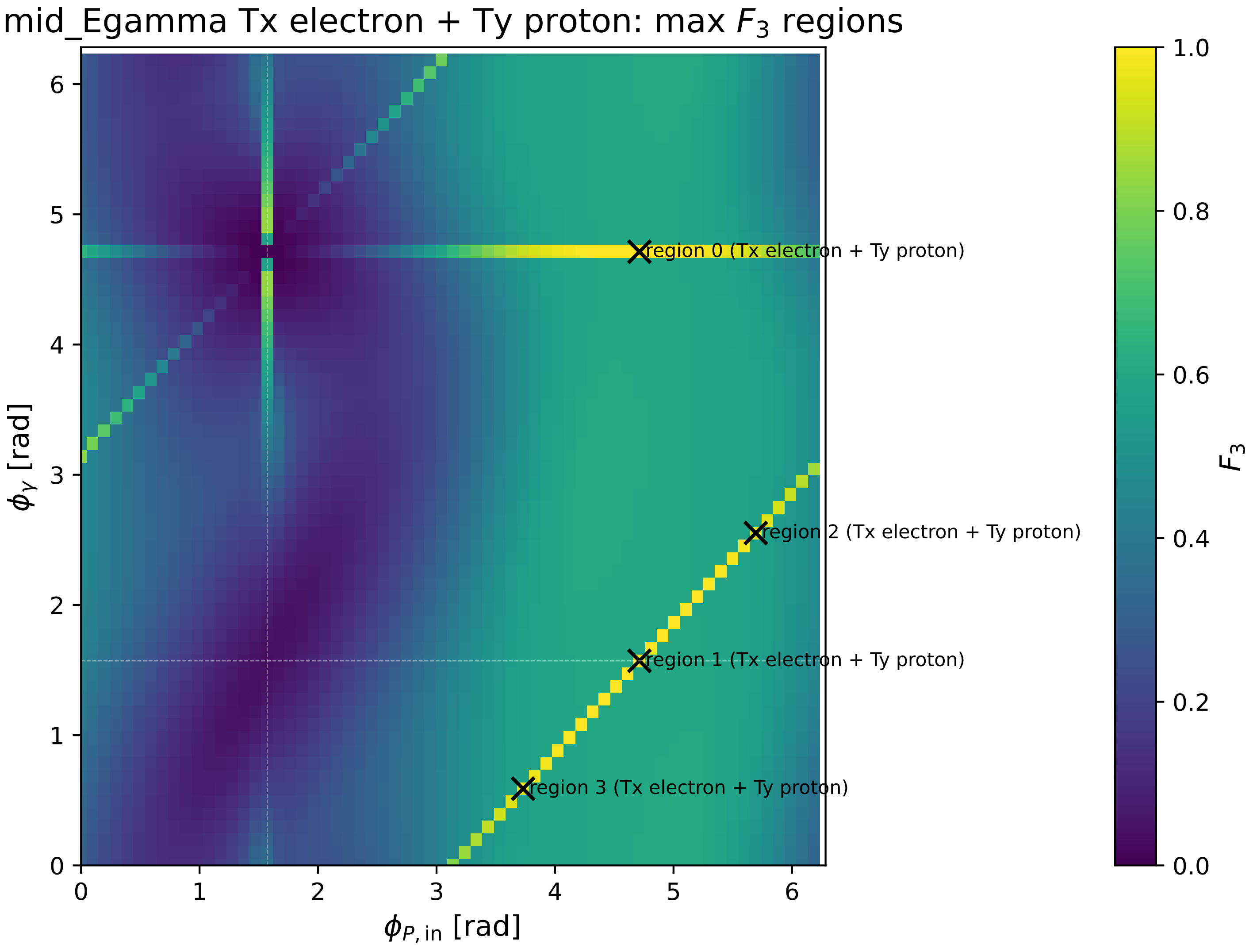
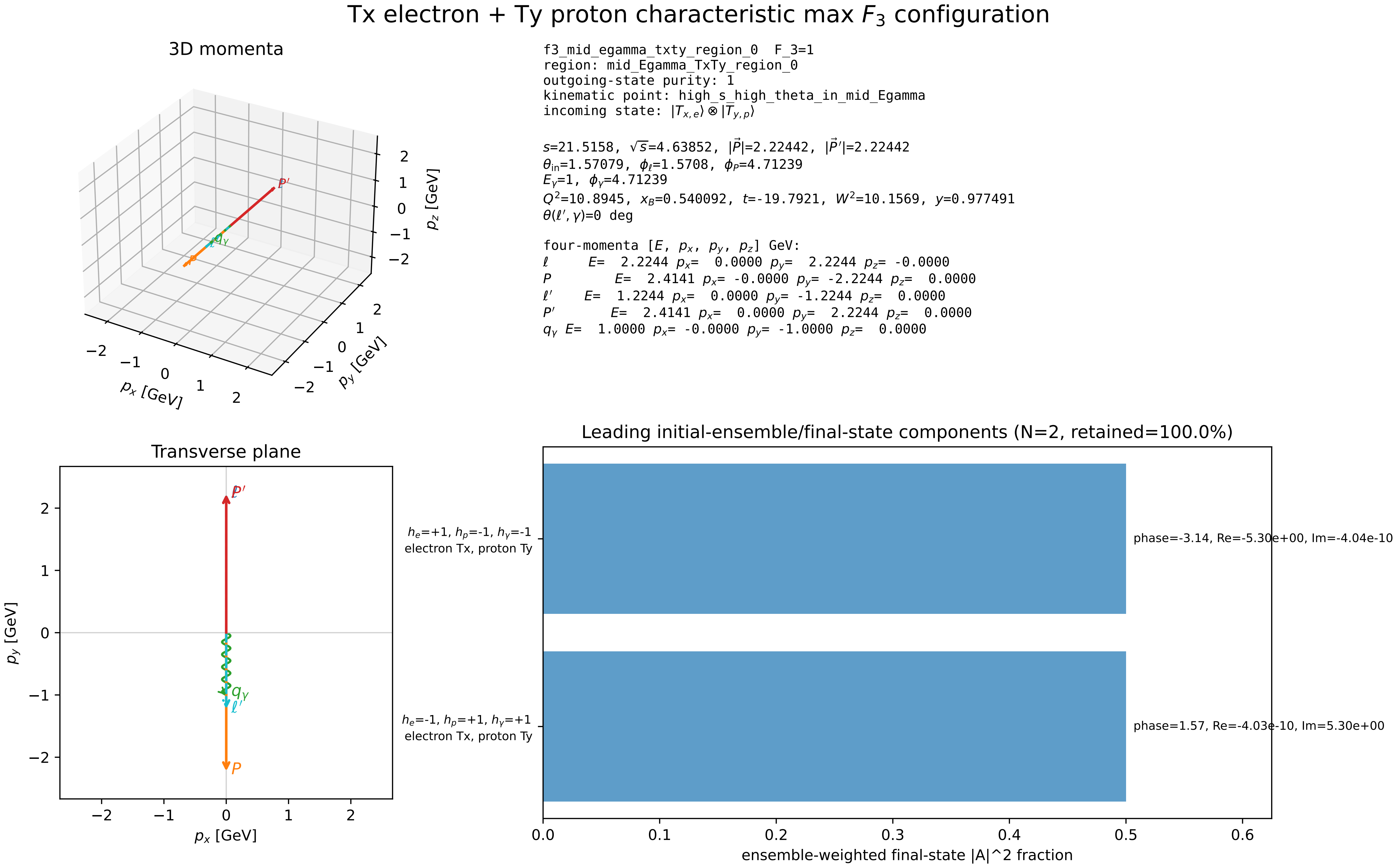


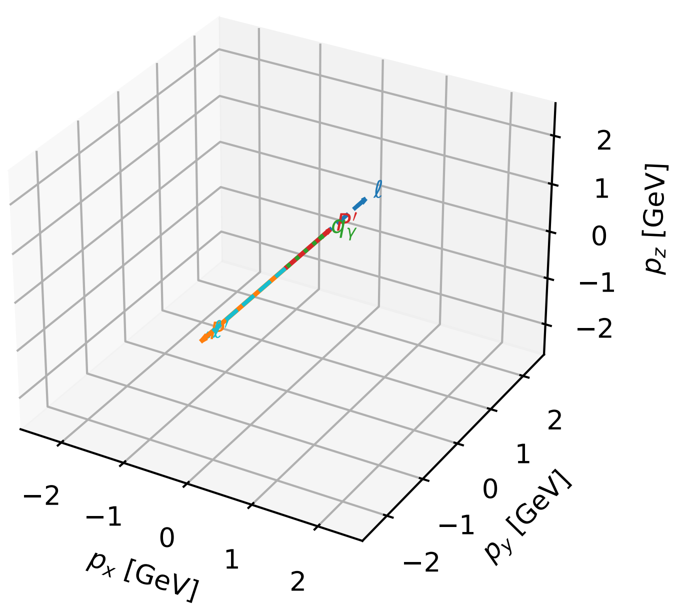
mid_Egamma Tx electron + Ty proton: max F_3 regions
region 0 (Tx electron + Ty proton)
region 2 (Tx electron + Ty proton)
region 1 (Tx electron + Ty proton)
region 3 (Tx electron + Ty proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
F_3
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0



Tx electron + Ty proton characteristic max F_3 configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P'
q_γ
ℓ'
P
f3_mid_egamma_txty_region_0 F_3=1
region: mid_Egamma_TxTy_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|T_{x,e}\rangle \otimes |T_{y,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.22442
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=4.71239
Q^2=10.8945, x_B=0.540092, t=-19.7921, W^2=10.1569, y=0.977491
$\theta(\ell', \gamma)$=0 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 1.2244 p_x= 0.0000 p_y= -1.2244 p_z= 0.0000
P' E= 2.4141 p_x= 0.0000 p_y= 2.2244 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.0000 p_y= -1.0000 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=2, retained=100.0%)
h_e=+1, h_p=-1, h_γ=-1
electron Tx, proton Ty
phase=-3.14, Re=-5.30e+00, Im=-4.04e-10
h_e=-1, h_p=+1, h_γ=+1
electron Tx, proton Ty
phase=1.57, Re=-4.03e-10, Im=5.30e+00
ensemble-weighted final-state |A|^2 fraction

Tx electron + Ty proton characteristic max F_3 configuration

3D momenta

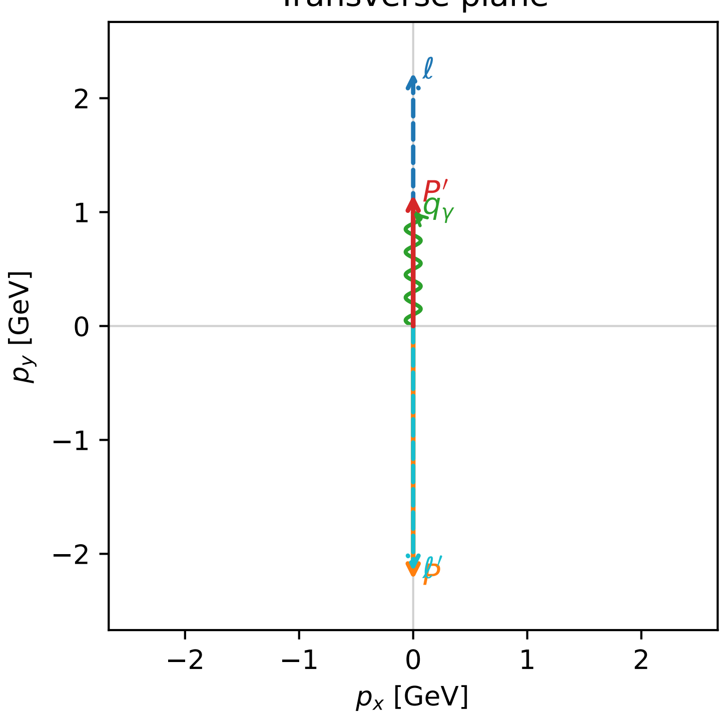


f3_mid_egamma_txty_region_1 F_3=1
region: mid_Egamma_TxTy_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|T_{x,e}\rangle \otimes |T_{y,p}\rangle$

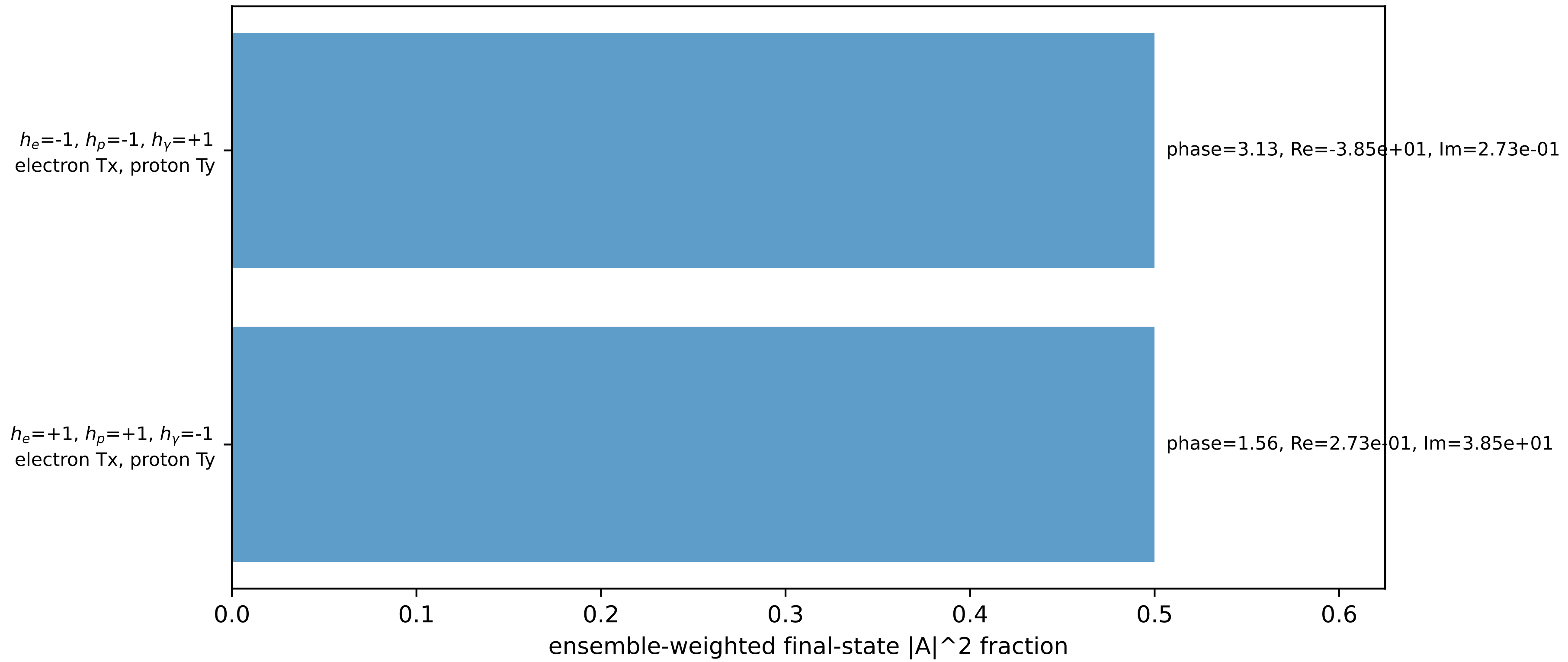
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.15253
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=1.5708
Q^2=19.1525, x_B=0.96635, t=-10.5424, W^2=1.54677, y=0.960429
$\theta(\ell',\gamma)$=180 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 2.1525 p_x= -0.0000 p_y= -2.1525 p_z= 0.0000
P' E= 1.4860 p_x= 0.0000 p_y= 1.1525 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.0000 p_y= 1.0000 p_z= 0.0000

Transverse plane

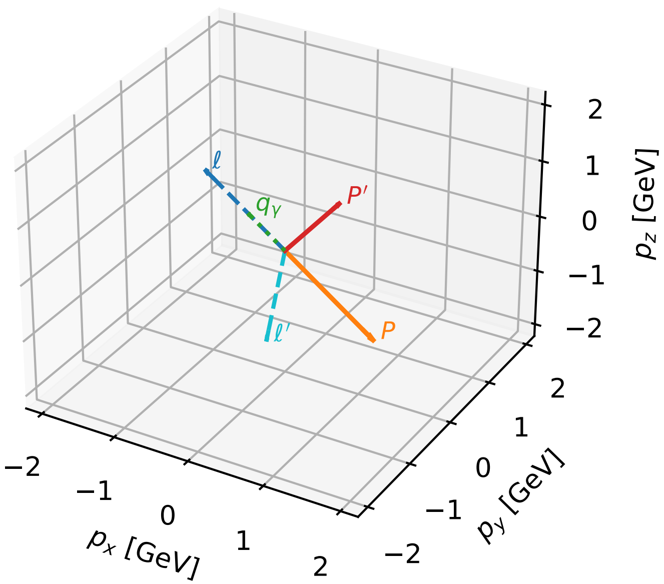


Leading initial-ensemble/final-state components (N=2, retained=100.0%)



Tx electron + Ty proton characteristic max F_3 configuration

3D momenta

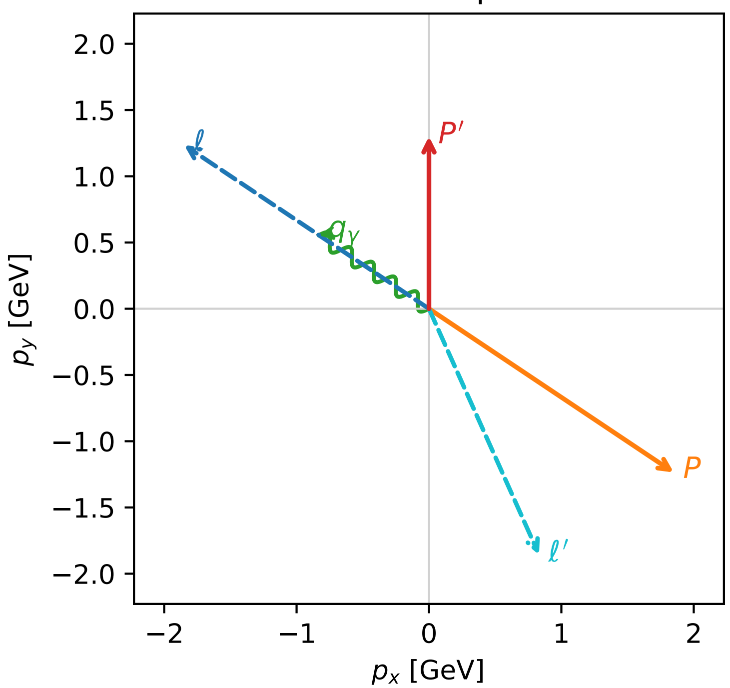


f3_mid_egamma_txty_region_2 F_3=0.967108
region: mid_Egamma_TxTy_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|T_{x,e}\rangle \otimes |T_{y,p}\rangle$

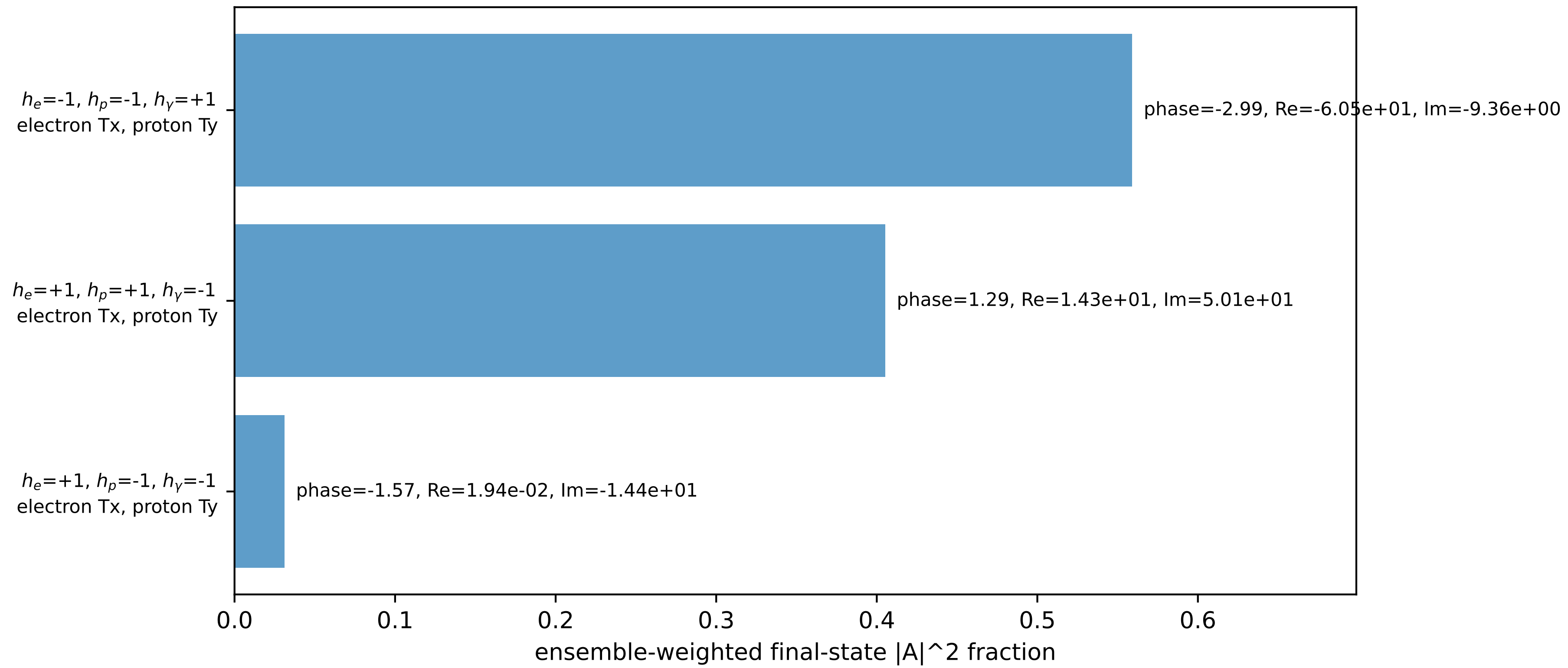
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.30122
$\theta_{\rm in}$=1.57079, ϕ_ℓ=2.55254, ϕ_P=5.69414
E_γ=1, ϕ_γ=2.55254
Q^2=16.7159, x_B=0.904629, t=-9.20119, W^2=2.64213, y=0.895436
$\theta(\ell',\gamma)$=147.873 deg

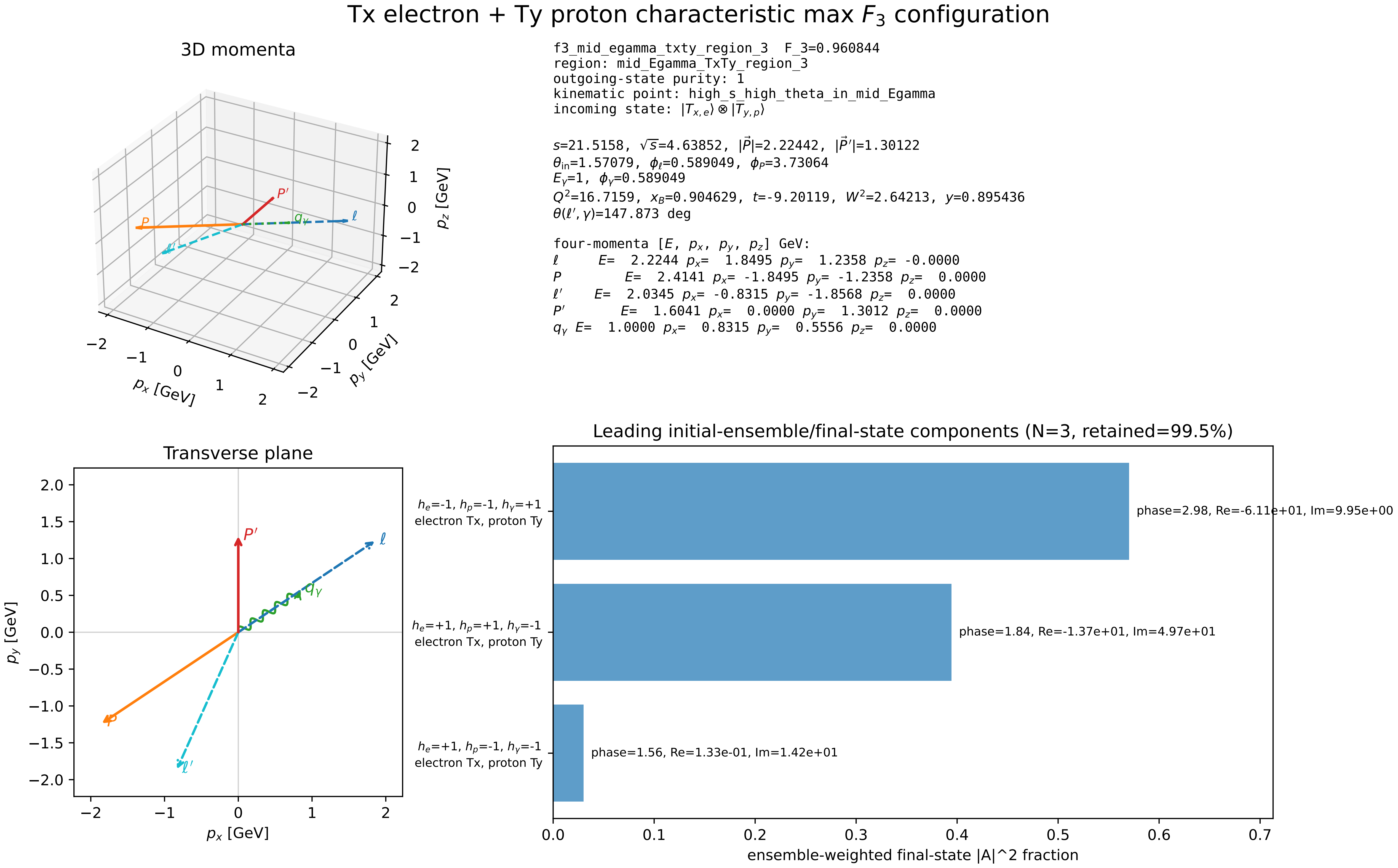
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -1.8495 p_y= 1.2358 p_z= -0.0000
P E= 2.4141 p_x= 1.8495 p_y= -1.2358 p_z= 0.0000
ℓ' E= 2.0345 p_x= 0.8315 p_y= -1.8568 p_z= 0.0000
P' E= 1.6041 p_x= 0.0000 p_y= 1.3012 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.8315 p_y= 0.5556 p_z= 0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=3, retained=99.5%)





Tx electron + Ty proton characteristic max F_3 configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
f3_mid_egamma_txty_region_3 F_3=0.960844
region: mid_Egamma_TxTy_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|T_{x,e}\rangle \otimes |T_{y,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.30122
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0.589049, ϕ_P=3.73064
E_γ=1, ϕ_γ=0.589049
Q^2=16.7159, x_B=0.904629, t=-9.20119, W^2=2.64213, y=0.895436
$\theta(\ell',\gamma)$=147.873 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 1.8495 p_y= 1.2358 p_z= -0.0000
P E= 2.4141 p_x= -1.8495 p_y= -1.2358 p_z= 0.0000
ℓ' E= 2.0345 p_x= -0.8315 p_y= -1.8568 p_z= 0.0000
P' E= 1.6041 p_x= 0.0000 p_y= 1.3012 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.8315 p_y= 0.5556 p_z= 0.0000
Transverse plane
Leading initial-ensemble/final-state components (N=3, retained=99.5%)
h_e=-1, h_p=-1, h_γ=+1
electron Tx, proton Ty
phase=2.98, Re=-6.11e+01, Im=9.95e+00
h_e=+1, h_p=+1, h_γ=-1
electron Tx, proton Ty
phase=1.84, Re=-1.37e+01, Im=4.97e+01
h_e=+1, h_p=-1, h_γ=-1
electron Tx, proton Ty
phase=1.56, Re=1.33e-01, Im=1.42e+01
ensemble-weighted final-state |A|^2 fraction